



TRT 101

TESTOSTERONE REPLACEMENT THERAPY - 101

“Dave has been an invaluable resource and blessing on my path to recovery from a major endocrine disruption. If you simply just want to optimize your full health or if you’re battling a difficult health condition, there is no better practitioner to help you reach your health goals.”

— Connor, Texas USA

DAVE LEE

Welcome to TRT 101.

This comprehensive, yet concise, actionable guide contains everything you need to know for beginning Testosterone Replacement Therapy (TRT).

This is the guide I wish I had over five years ago when I first began TRT, and contains a blueprint for TRT, that I constructed based on experience with over 500 clients over the last two years.

This is not a cookie cutter, one-size-fits-all protocol.

This is a roadmap designed to guide you with all the actionable tools and information you will ever need to make the right start with TRT.

The key areas covered are as follows:

- Should you start TRT?
- What is the best way to initiate and administer TRT?
- What kind of blood work should I get, when and what does it mean?
- And, most importantly, how to get TRT “dialed in”

***Note:** Hormone Replacement Therapy is a highly nuanced and specific field that goes well beyond the bounds of a 30 page pdf. For deep dives or expanded information on each topic, check out the TRT and Hormone Optimization channel on Youtube where myself and other experts in the field have answered many questions on various topics.*



TRT and Hormone Optimization - <https://www.youtube.com/c/FoxPowerBasement>

Many men on TRT will also require other forms of hormone replacement therapy, including thyroid, pregnenolone, DHEA, progesterone, cortisol as well as fertility protocols. These topics will be referenced here, but are covered in specific detail in my “Beyond TRT” guide, that will be released later this year.

CONTENTS:

CHAPTER I

TESTOSTERONE REPLACEMENT THERAPY -
WHAT IS IT? IS IT RIGHT FOR YOU?

CHAPTER II

THE RIGHT AND WRONG WAYS TO DO TRT

CHAPTER III

INITIATING TRT

CHAPTER IV

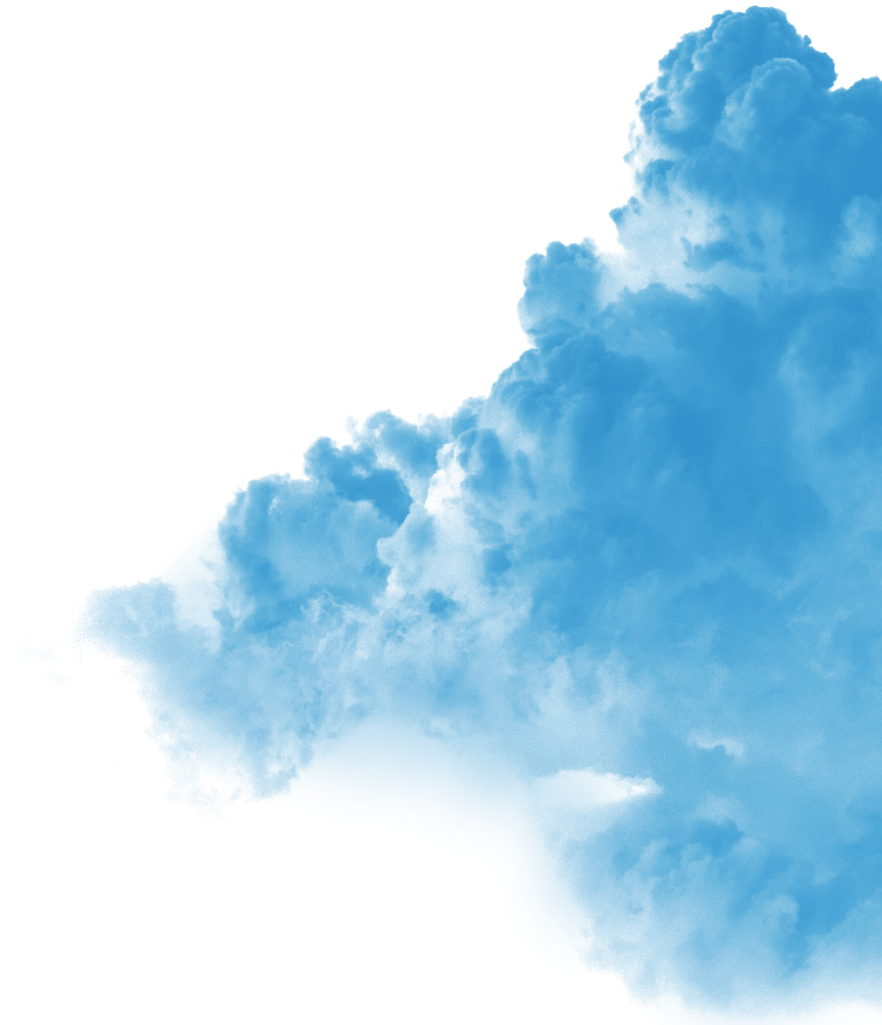
DIALING IT IN

CHAPTER V

FAQs

CHAPTER VI

AFTERWORD





CHAPTER I

TESTOSTERONE REPLACEMENT THERAPY - WHAT IS IT? IS IT RIGHT FOR YOU?

WHAT IS TRT?

Testosterone Replacement Therapy (TRT) is a form of “interventional endocrinology” where a man’s endogenous (internal) production of testosterone is replaced with an exogenous (external) form of testosterone.

This is initiated due to the endogenous level of testosterone not being optimal for the man, resulting in a variety of negative symptoms.

These negative symptoms are often amplified in intensity when the levels are lower, or the deficiency has been left untreated for an extended period. This is because low testosterone levels are detrimental to a man’s physical and mental health, and low testosterone levels are one of the causes behind many of the diseases of ageing in men.

TRT is an ongoing treatment, and involves shutting down the HPG axis, that is the signal between the brain and the testicles to make testosterone. As the body senses an external presence of testosterone, this creates a “negative feedback” system to the pituitary gland (the area of the brain responsible for signaling the production of testosterone, as well as other hormones), reducing the endogenous level of Luteinising Hormone (LH) down to zero. In turn, this lack of LH causes the testicles to go “offline” and stop producing testosterone.

And thus, a man’s testosterone is replaced.

Why?

Many men are not able to make optimal testosterone levels, due to a variety of factors. These include:

PRIMARY HYPOGONADISM: Failure of the testicles to produce optimal testosterone.

Diagnosed via elevated levels of LH - this is the pituitary gland sensing a lack of testosterone in the body and attempting to up regulate production, but the testicles are unable to respond.

Causes: Ageing (andropause), testicular injury, testicular cancer, orchiectomy.

Note:** It is often believed to this day, that Andropause (age related testosterone declines over a man's lifetime, men's equivalent of menopause) is the primary cause of low testosterone and that it does not occur in younger men. While this may have been true in previous years, in the recent decade **this could not be further from the truth.

? Should you commence TRT if you have primary hypogonadism?

Absolutely.

If the hardware is damaged, there is no way to increase natural testosterone production and an elevated LH level means your body has essentially already maxed out its natural production. If you have elevated LH levels and low T symptoms, **TRT is a must.**

SECONDARY HYPOGONADISM: Failure of the pituitary to signal the testicles to make optimal testosterone.

Diagnosed via suppression of LH - this is the body down regulating its production of testosterone due to stress.

Causes: Obesity, psychological stress, malnutrition, drugs and alcohol, chronic disease, auto- immune conditions, hypothyroidism, chronic pain,

history of anabolic steroid use (particularly when young and/or without proper Post-Cycle Therapy PCT), head injuries.

All of these are mediated via inflammation and the resulting down regulation of gonadotropins. The body prioritises survival over thriving and fertility in a chronic stress state. This is one of the consequences of chronic stress.

? Should you commence TRT if you have secondary hypogonadism?

This is less straight forward.

If the root cause of the pituitary suppression can be identified and resolved, that should be the first point of call. This is why it is important to work with a skilled practitioner and have an open, honest conversation about where you are at - so that your practitioner can determine if TRT is right for you.

While many of these states can be reversed... many can't be.

Time and time again, I see cases of these root causes resolving, but LH/FSH still remaining suppressed. This is very common in the case of PTSD, head injuries, or after long periods of caloric deprivation.

Many people are also struggling with the situation they are in, such as obesity or chronic disease, and would not be able to resolve the root cause issue without optimising their testosterone levels.

This is why secondary hypogonadism requires a case by case assessment.

? My question is - are you doing everything that you know you should be doing? Are you eating right, training, getting enough sunlight, sleeping well and looking after your physical and mental health?

If the answer is yes, and you are still suffering from secondary hypogonadism, then it is worth pursuing.

SYMPTOMS

The symptoms of low testosterone are very broad and extend to both acute and chronic effects of testosterone deficiency. This means that while low testosterone is problematic in the here and now, it can have devastating long term psychological and physiological effects on men.

The most common symptoms I see of low testosterone are: depression, anxiety, erectile dysfunction, low libido and fatigue.

However, it is important to note that a number of factors can cause these symptoms, and low testosterone itself can be a symptom of other problems (such as the factors mentioned above).

This is why it is extremely important for your practitioner to perform a comprehensive workup for both symptoms and blood work, to make sure that your symptoms are definitely coming from low testosterone.

 **Important note:** While testosterone will generally make most men feel “better” - if the problems are not stemming from low testosterone, testosterone will not fix the problem. While testosterone is beneficial for a variety of applications, it is not a panacea.

Younger men who have not had optimal testosterone levels during instrumental development years (late teens and early twenties) often have issues with learned fear (often diagnosed as generalised anxiety disorder or social anxiety disorder) and further develop secondary depression due to manifestation of this state. This **lack of androgenisation** also affects the physical development of the body, resulting in reduced muscle mass in response to adequate stimulus, as well as postural problems, metabolic syndrome (insulin resistance) and often, psychogenic erectile dysfunction.

Many of the secondary symptoms I see for low testosterone are the outcomes of an extended period in the low testosterone state, where the individual has lower motivation, lower threshold to stress and displays fewer masculine values and physical traits.

This can drastically offset a man's trajectory towards proper development in his career, physicality and understanding of himself.

The most common things I hear in consultations with clients who are describing their low testosterone symptoms are:



"I could go weeks without sex and not be phased."



"I can still have sex but my drive is very low, I could take it or leave it."



"I just want to crash on the couch when I get home from work every day."



"I always feel drained even though I sleep 8 hours a night."



"I have little enjoyment or drive for the things I used to enjoy."



"I'm easily overwhelmed by stress."



"I take days to physically recover from exercise."



"I wake up during the night and can't get back to sleep."



"I don't have the energy to do anything during the day but I can't sleep at night."



"No matter how much I train and eat right I can't build muscle."



"I've lost my masculine essence or 'mojo'."

The most common one that brings men to see me, outside the sexual issues, are the mental health and the physical recovery after exercise. In fact, it's often being completely wiped out for days after the gym that guys go "hey... this isn't all in my head, this definitely isn't right".

BLOODWORK

Blood work is almost never done correctly on TRT.

There are key markers that are not understood, overlooked or misinterpreted that put guys in the wrong direction time and time again.

Blood work is covered at various points in this guide for various stages of TRT.

I recommend checking the following blood values prior to starting TRT and having them assessed via an informed provider who can understand the relationship between each system in the body.

For example, if testosterone is suppressed, it is important for your provider to be able to look at other blood panels, or ask the right questions to assess WHY this may be, rather than just putting you on TRT because testosterone is low.



Please note: Depending on where you are in the world, your lab will measure in different units. I have included the most common units used in Australia and Europe. If your tests are not in the units you see here, there are plenty of online converters you can use to work it out.



TOTAL TESTOSTERONE:

An important measurement of how much testosterone your body is producing (or if on TRT, you are administering).

However, due to the pharmacodynamics of testosterone in the body (how it is absorbed and circulated), **total testosterone only paints part of the picture** and is **less useful than free testosterone** for determining if the individual is suffering from an androgen deficiency.

This is due to the actions of binding proteins Sex Hormone Binding Globulin (SHBG) and Albumin, that bind the vast majority of total

testosterone in the body. Although it does not render the bound testosterone completely inactive, free testosterone is a more reliable measurement of how someone “feels” relative to their testosterone levels.

Total Testosterone is measured in ng/dl and nmol/L primarily. The reference range is often around 200-800 ng/dl or 8-28 nmol/l in 2022.

However, in my experience, I have seen the reference range as low as 5-25 nmol/L and as high as 12-50 nmol/l in different labs, from different clients all over the world.

It is very well known that testosterone reference ranges are dropping all over the world. As testosterone levels fall in patients, averages and standard deviations are recalculated.

It is important to understand that **reference ranges have fallen significantly** in the last decade alone (my first reference range for total testosterone in 2016 was 10-31 nmol/l and most recently in 2022 it was 6/26 nmol/L) ...and we have only been able to measure serum testosterone levels for a small handful of generations.

It is hypothesised (with significant evidence) that testosterone levels have been falling for decades, particularly in the modern world post the industrial revolution, where our modernised diet has been significantly altered away from containing ample amounts of fat soluble nutrients, along with significantly more sedentary and polluted lifestyles. **For these reasons, it is unsurprising that many men need to go above the “reference range” to resolve their symptoms.**

This does not mean you have “supra physiological” levels of testosterone - as the average testosterone levels in 2022 simply does not equate to the physiological testosterone potential of man.



FREE TESTOSTERONE:

Free Testosterone is a measure of the bioavailable testosterone in your body at the given time you had your blood drawn and is most aligned with your symptoms of optimal or deficient testosterone.

Free Testosterone can be impacted by excessive levels of SHBG (discussed below) and is the primary target of TRT optimisation and the level to “dial in”.

Free Testosterone is often measured in pmol/L and the current reference range in Australia at the time of writing this guide is 200-600. I have seen it as low as 150-600, however, in 2016 the standardised reference range was 300-800 pmol/L. I have seen the range as wide as 100-1350 pmol/L. *This is important to understand, because a “mid- range” testosterone level in 2022 would have been considered “below range” and “hypogonadal” less than a decade ago.*

It is absolutely crucial that men understand that the reference range is NOT an accurate measure of where their testosterone should be to be an optimal healthy male - it is a measure of where the average male in society is today.

The Important Question: What level is “low” and at what level should you start TRT?

There is no set answer to this.

The action of testosterone in the body is dictated by many factors beyond simply how much there is in serum. (*Androgen receptor function, neurotransmitter genetics etc*). This means that one guy might feel great at a certain level, while another guy has negative symptoms at the same level.

It is important to understand that you do not need to be below the “range” to have low testosterone and to be suffering with hypogonadism.

I typically see clients having low testosterone symptoms with free testosterone under 400 pmol/L. The lower the levels, the worse the symptoms. However, it is important to understand that while you may not be suffering as much in the 300s as you would in the 100s - you can still benefit from optimising your testosterone levels with TRT. I would not be ruling out a low T diagnosis in the

400s, but the higher the free testosterone, the more important it is to check for other root causes.

While Total Testosterone is a factor, it is most often **free testosterone that is crucial in making the low T diagnosis**, as an individual with high total high SHBG and low free can have negative symptoms.

Most clients I see have a total testosterone under 20 nmol/l, with severe cases being below 10 nmol/l. The average testosterone level I see in young men with hypogonadism is around 15nmol/L, that is significantly higher than what many would consider "low".



SHBG:

SHBG is the main red herring I see men obsessing over.

No matter if it's high or low, people want their SHBG higher or lower and will pinpoint it for their negative symptoms.

While it has been shown in studies that low SHBG is associated (correlated) with diseased states, it is important to understand that SHBG is secondary to the problems caused by the diseased state, not the low SHBG itself causing the diseased state.

This means that low (or high) SHBG can be a symptom of a problem, but it is not the problem itself.

Low SHBG (usually single digits) CAN be caused by **hypothyroidism and insulin resistance**. *However, there are many people with optimal thyroid function and insulin sensitivity who genetically have low SHBG - so it is a signal for investigation but not cause for concern.*

SHBG is increased by time in ketosis. So, diets that deliberately induce prolonged states of deep ketosis (including OMAD style intermittent fasting), can drive SHBG up to a point it is problematic in some people. And this means that this way of eating is not agreeing with your biology.

SHBG also increases with age as one of the driving factors of low testosterone in elderly men, outside of testicular failure. High SHBG can also be a symptom of liver disease. **However, as with low SHBG, many healthy young men have high SHBG levels with no problematic root cause.**

I encourage people not to worry about SHBG.



ESTRADIOL:

Estradiol before initiating treatment is not an overly useful marker.

It can be worth measuring as if it is undetectable it can be causing problems with the joints, cognition or insulin sensitivity, while if it is elevated (in the context of low testosterone) it can indicate inflammation or stress related issues that may be able to be mediated to recover natural testosterone production.



PROLACTIN:

Prolactin in men is elevated by stress, as well as pituitary tumours.

In the event of a pituitary tumour (non cancerous) diagnosed via an MRI, cabergoline or pramipexole may be used to suppress prolactin down to a healthy level (bottom half of the range), and to hopefully eliminate the tumour over time.

Prolactin in men will suppress dopamine that will then impact the gonadatropin releasing cascade, causing low testosterone levels as well as symptoms of low dopamine, that have strong crossovers with low testosterone.

Prolactin can be elevated within or just beyond the range by psychological stress, chronic pain, hypothyroidism or metabolic syndrome.

However, if the prolactin is significantly elevated (double the top of range or beyond,) a pituitary MRI is absolutely warranted.

Many will opt to get a pituitary MRI if prolactin is even slightly elevated prior to starting TRT and this is an excellent idea to rule out a potential root cause for low testosterone.



PREGNENOLONE:

Pregnenolone is an important neurosteroid for a variety of cognitive and physiological functions.

(I have an in-depth lecture on pregnenolone on YouTube via the TRT and Hormone Optimisation channel and it will be covered in depth in "Beyond TRT".)

If there is a history of head injury or PTSD, it is worth checking for pregnenolone levels, as a pregnenolone deficiency will cause negative symptoms (anxiety, poor memory, stutter, depression, anhedonia) that cannot be resolved by anything but pregnenolone.

Pregnenolone deficiency will also cause side effects from TRT and is worth checking either prior to, or after initiating, treatment if there are negative symptoms or a potential root cause for deficiency.



PROGESTERONE:

Progesterone is often touted as a "female hormone".

It is very rare for a man to have a progesterone deficiency however, progesterone has important GABA (calming) properties, and a deficiency can predispose an individual to insomnia and anxiety.

Progesterone can usually be optimised via pregnenolone supplementation, but this is case by case.

A varicocele (swollen veins in the scrotum) can cause elevated progesterone that will inhibit 5 alpha reductase and thus reduce DHT levels. *This is why it is important to check progesterone in the presence of a varicocele as it can support the decision for either a varicolectomy or TRT.*

It is commonly reported that varicoceles do NOT lower testosterone levels. *This could not be further from the truth.*

Varicoceles (even unilateral) will cause dysfunction of the Leydig cells (where testosterone is produced in the testes) and the altered ratio of progesterone to testosterone has an anti-androgenic effect.



DHEA:

DHEA was once assumed to only be a precursor hormone with no intrinsic activity itself, but this has been debunked in the recent years.

DHEA has been found to be an important driver of a number of biological functions, particularly relating to metabolism, cognitive function and the immune system.

Low DHEA will cause issues with reduced penis sensitivity, delayed orgasm and low mood. *In fact, correcting low DHEA will often resolve mood issues if they are present.*

Low DHEA and low testosterone share many common symptoms, as DHEA is crucial for libido and sexual function, and is important for dopamine and serotonin production. It actually shares a lot of similarities with caffeine.

I find that DHEA below 4 umol/l can cause negative symptoms. More DHEA is not always better and is discussed in detail in my YouTube lectures as well as in "Beyond TRT".



TSH:

Unfortunately, many practitioners will check TSH by itself to determine and diagnose thyroid function. This is impossible.

An elevated TSH often suggests a cellular thyroid deficiency. *However, a low TSH (optimal is under 1 or close to) could mean there are secondary issues causing thyroid dysfunction and a full assessment of T3/T4 is needed.*

**T3:**

T3 is the primary active thyroid hormone in the body, discussed at length in Beyond TRT.

It is important to understand that optimal thyroid function is crucial for optimal metabolic health and a deficiency in thyroid is often the missing piece of the puzzle left unresolved for many on TRT.

Jay Campbell has spoken extensively about the importance of thyroid optimisation and it being something “left on the table” for many men looking to optimise their hormonal health.

T3 heavily governs energy metabolism.

**T4:**

T4 acts primarily as a precursor to T3.

However, T3 cannot cross the Blood Brain Barrier (BBB) and the brain requires T4 to cross the BBB to be metabolised into T3.

**THYROID ANTIBODIES:**

Elevations in thyroid hormone suggest hashimotos thyroiditis, and warrants further treatment and investigation.

**IRON PANEL:**

Anaemia is less common in men than women (as men do not bleed monthly and tend to consume more dietary heme iron due to higher caloric intake) but it is important to note that anaemia can cause many symptoms of low testosterone.

**LIVER PANEL:**

NAFLD (Non Alcoholic Fatty Liver Disease) is present in many patients with low testosterone and other endocrine dysfunctions, whether as a root cause or complication of chronic hormonal deficiency.

It causes symptoms of malaise, as well as many low testosterone symptoms and will also contribute to hypogonadism.

Unfortunately, NAFLD is not regarded or treated by western medical doctors as a whole, as it is not treated with pharmaceuticals, and the recommendations is often to “just lose weight”, if anything.

If Bilirubin is elevated, it is crucial to have a liver ultrasound. An individual may have Gilbert syndrome (elevated bilirubin in the absence of a fatty liver) and can be ruled out via ultrasound.

NAFLD is extremely common in people who have been overweight for extended periods.

Other forms of liver dysfunction that are more serious can also contribute to hypogonadism, which is why it is important to keep an eye on liver health.

The health of the liver is also crucial to the success of TRT.



KIDNEY PANEL:

Renal function should always be monitored, especially when exogenous hormones are being utilised. This is not because exogenous hormones damage the kidneys, but the kidneys are integral in supporting the metabolism of these hormones.



CHOLESTEROL PANEL:

I suggest looking into the works of Paul Saladino and Dr Ken Berry regarding HDL and LDL levels.

I primarily focus on triglycerides as markers of inflammation, and will focus on HDL/LDL when triglycerides are elevated and there are other risk factors for cardiovascular disease, such as elevated CRP or A1C.



CRP:

An excellent marker for full body inflammation. This marker can be influenced by injuries or recent training, so it is important to have blood work when fully rested and recovered.

Elevated CRP suggests investigation elsewhere for the root cause of the inflammation.



A1C:

A useful measurement for 24 hour blood glucose variation. I find this to be the most valuable marker when assessing insulin sensitivity, followed by fasting insulin.



25 HYDROXY VITAMIN D:

Vitamin D plays an essential role in the endocrine system. However, Vitamin D is not the only important nutrient the human body gets from the sun, but it can be a proxy for adequate sun exposure and nutrition.



CHAPTER II

THE RIGHT AND WRONG WAYS TO DO TRT

While there is no “right” way to do TRT, there are definitely wrong ways to do TRT. *Let's get them out of the way first.*

1 PELLETS, GELS, PATCHES AND (NON-COMPOUNDED) CREAMS

None of these options works.

Pellets have a risk of infection, are expensive, suck to get put in and make it virtually impossible to titrate the dose. The dose is almost always too low, and severely hypogonadal by the time the pellets are refreshed. Pellets work well for cattle, not humans.

Gels are often 1%, meaning you will need to slather your entire body in the goo to get a decent amount of testosterone. This is extremely cumbersome, expensive and has a high risk of transference. Also, its low bioavailability means inconsistent, sub optimal results.

Patches sound great on paper, but do not deliver anywhere near enough testosterone for them to be effective, and can cause reactions at the site of the patch. *Patches are good for nicotine, not testosterone.*

Compounded creams work well (discussed below), but commercially available creams (androforte, for example) do not. This is because their concentrations are too weak and they are not enhanced for optimal absorbency, meaning that you simply cannot reach optimal levels by using them.

Some guys may “get away” with androforte 5% but would be much better off, in terms of cost and convenience, using a 20% compounded cream at the same dose.

2 TESTOSTERONE UNDECANOATE (REANDRON, NEBIDO)

Testosterone Undecanoate is an example of “if it sounds too good to be true, it probably is”.

Testosterone Undecanoate promises TRT with only 4 injections per year, with a long acting formulation allowing for gradual absorption over a 3 month period. *Unfortunately, in reality, this does not eventuate.*

However, patients are put on these protocols all over the world every day, and it is even considered the standard of care by many endocrinologists, including in Australia.

The half-life of undecanoate is simply too short to allow for such infrequent injections.

? Is there still testosterone in the system after 12 weeks?

Sure. But nowhere near enough to be optimal ... and for most, it dips back down to as low, or even lower than they were before they started.

The other issue with Testosterone Undecanoate protocols (1000mg per shot) is that the amount of testosterone is far too low, even if it were to be evenly distributed over the period.

Depending on the protocol, most guys end up on the equivalent of around 100mg a week of testosterone enanthate or less which, unless you're a hyper responder, is too low to achieve optimal testosterone levels.

Note: *Testosterone Undecanoate could potentially be injected weekly or fortnightly to allow for stable levels. However, it is almost never prescribed this way and patients are often required to have their injection done at their doctor's office.*

Better than nothing for some, but far from ideal.

3 INFREQUENT ADMINISTRATION OF TESTOSTERONE CYPIONATE OR ENANTHATE

Testosterone Cypionate and Enanthate are currently, in my opinion, the best option for the vast majority of patients on TRT.

Testosterone Cypionate and Enanthate are the two most common forms of injectable testosterone worldwide for both prescription and performance enhancement. This is due to their ease of availability and moderate half-life.

Trust me - if there was an option for TRT that delivered optimal results that didn't involve multiple injections per week we would all be doing it.

However, when physicians receive instruction on prescribing testosterone enanthate or cypionate, the guidelines are 1 injection of 250mg every 2-3 weeks. *This is because in studies, it is shown that the testosterone is active in the body for this amount of time.*

The problem is that the medication is not evenly distributed over this period. This is not how half-lives work. *Caffeine has a biological half-life of 8-10 hours, but we all know a cup of coffee does not have a gradual, even release over 8-10 hours.*

Testosterone cypionate and enanthate have the same advertised half-life of just over a week, that varies depending on which publication it is.

However, in my opinion, based on my understanding of what a "half-life" is, I believe it is closer to 4 days.

In my experience dealing with hundreds of clients on these formulations, as well as myself, most people have a spike on day 1 or 2 after an injection, and begin to feel diminished effects shortly after, This varies with the individual.

Some guys seem to metabolise Testosterone Enanthate/Cypionate very quickly and require daily administration to feel optimal.

While some guys can get away with weekly injections or every 5 days, in my experience, twice weekly is the minimum injection frequency to feel optimal on Testosterone Cypionate/Enanthate for the average person.

(Yes, there are outliers on either side.)

Most doctors will prescribe every fortnight, or once a week. For the majority of people, this will cause a peak at the start of the week and a trough at the end of the week, where you will feel drastically different.

This high and low swing on a weekly basis can cause mood dysregulation, as well as other symptoms of hormonal fluctuations like acne, water retention and a red face. *Watch out for this. If your doctor is not prescribing testosterone more frequently than weekly, it's a red flag that s/he's not up to date with best practices.*

4 AROMATASE INHIBITORS

In "Beyond TRT" I expand in detail on estrogen, as well as "estrogenic side effects" and how to resolve/understand these issues fully.

Any practitioner prescribing an aromatase inhibitor to initiate TRT should, in my opinion, be avoided.

While it was standard of care to use aromatase inhibitors around the time that I started TRT back in 2016, many practitioners and experts in the space have come to understand the role of estradiol and aromatase in men, and see estrogen in a less stigmatised way.

Aromatase Inhibitors likely migrated from the body building culture, where they are used to suppress estrogen production via disabling the enzyme that is required to make it.

This allowed body builders to run “supra physiological” levels of testosterone with fewer side effects.

However, in recent years, as more men have started to understand the benefits and role of estrogen in the body, as well as the harms of suppressing aromatase systemically, many practitioners have dropped the use of aromatase inhibitors and have found their patients have had better results across the board.

This is a long and complex topic so I will briefly outline the rationale for why aromatase inhibitors should be avoided while on TRT.

- **Early research into the harmful effects of estrogen was not able to distinguish between endogenous estrogen and xenoestrogens** (harmful estrogenic compounds in plastics, pesticides and pollution)
- Many studies on the **harms of estrogen are not done in men with high testosterone levels.**
- **It is unlikely, based on the current evidence, that endogenous estradiol (E2), is responsible for the negative side effects it is attributed towards,** which is why lowering estradiol via aromatase inhibition solves some problems but creates new ones
- From my research and observations, **the negative side effects experienced on TRT that men blame estrogen for are to do with inflammation,** which in turn drives up estrogen production, as estrogen is increased by stress and has anti- inflammatory roles (similar to cortisol).
- Men with liver issues and metabolic issues will have issues with regulating and metabolising estrogen (liver metabolises E2 into E1 for elimination, T3 regulates production of aromatase and suppresses cortisol production).
- Aromatase is expressed in fat tissue, and **many men with side effects on TRT are >15% body fat.**

The symptoms associated with “high estrogen” should be instead viewed as symptoms of excessive inflammation and/or metabolic disease.

Many doctors will put men on an AI prophylactically (preventatively) to avoid these “problems”, or will use an AI as a panacea for TRT side effects.

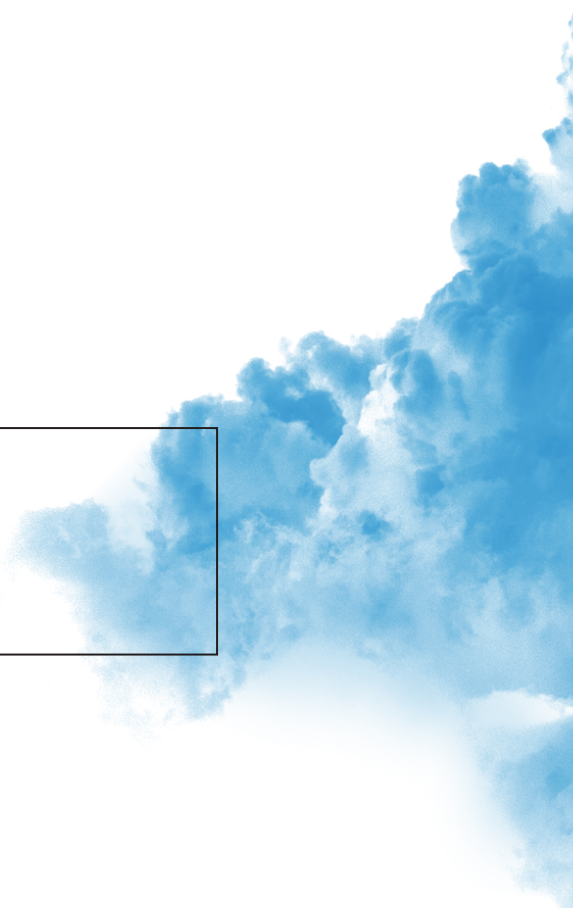
Estrogen is a paracrine hormone that is made on demand by the body locally in the tissue. Shooting for a target level of estrogen in serum is a fool's errand and shows a lack of understanding of estrogen itself. Avoid.

5 COOKIE CUTTER PROTOCOLS

The classic “Testosterone, HCG and AI” protocol is offered by many clinics as an opportunity for them to triple their profit per customer.

Creating problems you didn't know you had and selling you the solution is a classic marketing technique that is used heavily by practitioners who sell cookie cutter protocols.

Each TRT protocol needs to be customised for the individual. Many practitioners will have similar approaches or ways of addressing certain situations, but practitioners need to understand the importance of, and how to customise, the treatment for each individual, based on their unique circumstances.



CHAPTER III

INITIATING TRT

From here on, I will be guiding you on how to start TRT. Please keep in mind this is not medical advice and is for educational information only. All examples are hypothetical and you must discuss any treatments with your doctor. Possession of testosterone without a prescription is considered an offence in many countries. Do not break the law.

FIRST AND FOREMOST: SET REALISTIC EXPECTATIONS

TRT is not Adderall and it's not the limitless pill either.

It's not an anabolic steroid cycle and it will not turn you into Superman. TRT is about unlocking your genetic potential as a man.

Testosterone builds the foundation that you currently do not, or never have had, that allows you to put in the work to become the man you want to be.

In a low testosterone state, men have low resiliency and are highly susceptible to stress.

In a high testosterone state, men are more resilient and have a higher threshold for stress.

This allows men to take on more challenges and have the drive and passion to become more than they are. However, it won't get you up off the couch. *And you won't become the man you want to be unless you build on the foundation. TRT will not solve your problems, but it will give you the potential to become the man who can.*

SECONDLY: UNDERSTAND THE RESPONSIBILITIES

Shutting down your body's endogenous testosterone production is the #1 side effect of testosterone. This carries a number of side effects that all men must understand fully before commencing TRT. Many practitioners do not explain these well, or at all.

TRT should be seen as a commitment for life.

Many low T men are indecisive and have a high level of fear of the unknown, that scares many men away from the treatment they need. While this should be taken seriously, it should not be a deterrent for those who truly need testosterone replacement therapy.

TRT creates an acquired hormone deficiency, meaning a man will be dependent on exogenous testosterone to function, the way a diabetic is dependent on insulin.

Testosterone is also a scheduled medication, meaning its access will be limited and controlled by your pharmacy. This means extra care and attention must be made to ensuring your supply of testosterone does not run out. *Where it is legal and safe to do so, I recommend men always procure a back up supply of testosterone to have in circumstances where there could be delays in re-ordering or picking up their prescription, or in the event of a circumstance where they are unable to access a pharmacy.*

Remember, TRT is for life, and sometimes things happen. And when these things happen, you don't want to be low T.

FINALLY: CONSIDER FERTILITY

 *Does TRT make you infertile? No.*

 *Does TRT impact fertility? Yes.*

TRT shuts down FSH that is crucial in the production of sperm. *However, not all men are infertile on TRT.*

If TRT acted as contraception, we would have fewer body builders causing accidental pregnancies.

A key factor in the misunderstanding around TRT causing infertility, is that many men are infertile prior to going on TRT (hint: low testosterone and

infertility often go hand in hand - where there's smoke, there's fire) and don't realise it until after they initiate treatment and then blame the TRT.

While many men, including me, have not had their fertility impacted, significantly after more than 5 years on TRT, the outcomes for longer periods, or individual variance, are completely unknown.

It is crucial to see TRT as a potential complicator for fertility and depending on how much of a priority future fertility is, I highly recommend taking into consideration the section on HCG later in this guide, as well as the fertility section in Beyond TRT.

Note: *If you're in your twenties, or even early thirties, and don't want to have kids I highly suggest being open to the possibility of changing your mind later in life. Most low testosterone men can't imagine having kids because life with low testosterone is already stressful. I was vehemently against having children for up until a few years into my treatment, and changed my tune to it being something I want for my future, and I have seen this happen many times. A decade or so on TRT will change you into a different man from the inside out, and what you want after that may be very different to what you want now.*

With all of that said, if you can check every item on this checklist, move ahead with the later steps in this guide.

- I have the symptoms of low testosterone, as well as lab work that indicates a deficiency.
- I have taken every step possible to maximise my own natural production and addressed any other health issues OR I am unable to resolve the issues that are causing low testosterone.
- I understand that I will be creating a dependency on TRT and this is a lifelong commitment.
- I understand that this treatment may complicate my future fertility.

Let's begin.

1 FIND A PROVIDER

Finding a provider is more or less straight forward depending on where you live. I have included a list of recommended providers for Europe, UK, Australia in the FAQs section of this book.

If you are having trouble finding a provider for TRT, I recommend ringing your local compounding pharmacy and asking for a list of doctors who prescribe testosterone near you.

You will be able to vet your provider based on the recommendations in the next section. However, I recommend asking around in groups or communities, or from friends or colleagues receiving treatment for a recommendation. It is very difficult to convince a doctor to do TRT the right way and you will have much fewer headaches.

2 START YOUR TREATMENT

I recommend initiating treatment with Testosterone Cypionate or Enanthate at a dose of 100mg - 200mg per week, split into a minimum of two injections, as a shallow, intramuscular injection.

Most people will opt to inject Monday and Thursday, but you can do another combination of days if you wish.

Note: *To split your dose exactly evenly over the week, you “should” inject Monday morning and Thursday night. However, this is not needed, as we are merely splitting up the dose to reduce the peaks and valleys - to equate to the same dose over the week. Therefore, it is not overly important to time this perfectly at 3.5 days, both in the morning or at night is fine.*

The dose you start with will depend on your circumstances and your provider.

I will start my clients with very low SHBG on a lower dose (100mg-120mg) while clients with higher SHBG I will start with a higher dose (closer to 200mg, depending on the levels).

You can start high and work down, or you can start low and work up, the end destination is the same.

You will be having a review of your bloods and symptoms shortly after commencing treatment, so as long as you are going to comply with your protocol and be consistent, any place in this range is a good place to start.

I prefer to start most clients on 150mg a week. This is because it is easy to split into 2 shots (0.3ml per shot) or 3 shots (0.2ml per shot) - assuming we are using a standard 250mg per ml preparation.

What to expect:

Some users will report a "honeymoon" period of 1-2 weeks, or even a bit longer, after commencing TRT.

While some put this purely down to placebo effect (always a factor) I do not believe the placebo effect is solely at play. I believe this is due to the increased transmission of dopamine from testosterone, as well as enhanced activation of the androgen receptor in the brain, resulting in an "enhanced" response to treatment.

Some find these first few weeks euphoric, others may find it over stimulating or anxiety provoking. *Others might not notice anything at first.*

All of these outcomes are normal. If you feel over stimulated - move.

Your body has some big adjustments to make, as your entire biology has

been forged and expressed in a low testosterone environment, and there are drastic changes at play, similar to when a child first enters puberty.

You may have some water retention, or you may have to learn to deal with a less agreeable spark or a “temper” you have never had to regulate. Your appetite may crank up, and you may get some nipple sensitivity. All of this is very normal and part of your body adjusting to the new level of testosterone.

I do not recommend being reactive during this period.

Trust the process, review your blood work at the advised follow up period (8-12 weeks after commencing) and start doing the work on doing all the things that will make you into the man you want to be.



Hint: Lift weights, do cardio, learn martial arts, learn meditation, practise cold/heat exposure, listen to audiobooks and podcasts.

How to do your first shot:

Injecting yourself is extremely unnatural, especially if you are not familiar with needles.

There is an easy, but still significant, learning curve when it comes to learning to self - inject anything, especially testosterone that is in an oil, making it thicker and more cumbersome to inject than water based formulations.

I recommend using 27g-29g, 1/2” insulin syringes, and, when starting injecting into the lateral delt. Once you get the hang of injecting with one hand, move to the ventro glute if you wish.

The 27g needle is thicker, the 29g is skinnier (higher number, skinnier needle). These syringes are designed for insulin (water) so testosterone

flows through these a little slowly, but it is worth being patient to save on wastage (all-in-one syringes have no dead space) as well as less pain and scar tissue. Remember, you're doing this for life, with a minimum of around 100 injections a year.

Make sure to get a 1/2" needle (smaller insulin syringes are only 8mm) which will allow you to hit the deltoid.

If you are particularly obese and lack delt development, you may need the next size up, a 16mm 27g needle, that comes as a separate attachment for a standard syringe barrel. Just make sure to use a barrel 1ml or less to be able to accurately measure your dose, but insulin syringes are king.

Drawing from the vial and injecting should take a total of around a minute or so, once you get the hang of it.

Don't worry if you're uncomfortable or nervous on your first injection, this is normal.

I recommend the delts over the quad, as you're only injecting a small volume of oil and you don't run the risk of hitting a nerve or artery - *that are as fun as they sound.*

Your first shot will probably give you a dead arm and will ache for a number of days. *My first shot in the glute made me feel like I had been shot for almost a week.*

Your muscle has not had oil injected before and produces an enhanced inflammatory response. Don't worry. *Your body will get used to it and you'll feel nothing following an injection after a few shots.*

Injecting becomes like a slightly less convenient version of brushing your teeth once it becomes part of your lifestyle. The needles are tiny. Don't worry about it.

Questions about subcutaneous administration as well as starting with a topical cream can be found in the FAQs section.

3 FIRST REVIEW

It is extremely important that you review your testosterone levels and your symptoms soon after beginning treatment.

It takes at least 6-8 weeks for levels to fully build up in your system.

Your first review is to make sure that you have not hyper (over) or hypo (under) responded to your treatment.

Each practitioner has different targets they roughly aim for on blood work, based on what they see as producing the best results for their patients.

Some prefer to keep their patients within the reference range, while some will understand that many, if not most, particularly younger patients will require higher levels than the reference range to achieve symptom resolution.

It is also important to understand that numbers fluctuate drastically, even on daily injections (which is the most stable TRT protocol available). **I often see numbers fluctuating by 20-30%, even on daily injections for long term clients.** This is because there are a number of factors that influence the speed by which the body metabolises the Enanthate or Cypionate ester, making the exogenous testosterone bioavailable in the body.

This is why both symptoms and levels are important.

Blood work is a guide to make sure you are in the ballpark and symptoms are key in making sure your TRT is optimal.

Keep in mind that early into treatment, while levels may have saturated, you do not feel the full effects of TRT at this point, not even close.

Just like puberty doesn't completely eventuate and manifest in a few weeks, testosterone optimisation therapy takes time to achieve its goal, which is masculinisation of the man, including mind and body.

This is why it is important to understand that at your first follow up, it is only expected that you will feel like you have taken a step in the right direction.

If you're 10-20% better than when you started, that's a good sign.

Remember, the main benefits of TRT take time, and the goal of this appointment should be to make sure you're close to the bullseye and then to give it time and let it settle.

I advise my clients to wait at least 12-16 weeks after this appointment for their next follow up. During the period between the first and second follow up, after making any adjustments, I again encourage my clients to do all the things they know they “should” be doing, and encourage them to really focus on doing the “work”. As I said previously, testosterone is the foundation, and many guys will sit around and wait for the house to build itself. It never does, and no matter how much foundation you put down (or how much testosterone you put in), you’ve still got to build on top of it.

4 SECOND REVIEW

At the first review, I advise guys to only check their testosterone levels - free and total (as well as SHBG that is used to determine the free in countries where free is calculated).

This is because early into treatment, other health metrics (the panels ordered prior to starting treatment) are still going to be in flux, and taking a snapshot of things in motion does not provide actionable information.

At this point, which is 4-6 months into treatment, I like to check a full comprehensive metabolic panel, that includes all the tests measured in chapter 1.

The panel I typically order for my clients at this stage, or whenever I require comprehensive analysis is as follows:

-  *Testosterone*
-  *Free Testosterone*
-  *SHBG*
-  *Estradiol*
-  *IGF-1*
-  *Prolactin*
-  *Progesterone*
-  *DHEA*
-  *Cortisol*
-  *Insulin*
-  *HBA1C*
-  *CRP*
-  *Liver Panel*
-  *Kidney Panel*
-  *Cholesterol Panel*
-  *Iron Panel*
-  *White Blood Cell Panel*
-  *Red Blood Cell Panel*
-  *25 Hydroxy Vitamin D*

 ***I like to compare side by side, the results from your before and after blood work to ask the question - are you objectively healthier than before you started?***

This is what your practitioner should be looking for at this stage, to make sure you are tolerating your treatment well, and it is benefiting you objectively as opposed to just subjectively.

Example: *Many TRT patients believe that TRT should be purely based on how someone “feels” and if they feel good, it must be working. This is nonsense - as many people would feel great on 500mg or more of testosterone, but this is not objectively good for their body long term, as it is supra physiological (beyond what the body could naturally make). This is why we must factor in both lab work AND symptom resolution in optimising TRT.*

I hope to see a reduction in CRP, Triglycerides, as well as fasting insulin (if they were elevated previously). I also expect to see a reduction in cortisol (if it was elevated) and no issues in liver, kidneys, RBC or WBC.

Note: *It is common to have a transient rise in Hematocrit, a measurement of blood viscosity when starting TRT. This is one of the reasons I don't recommend measuring it immediately after starting treatment as your body is adjusting. Elevated hematocrit results leads many doctors to recommend phlebotomies (blood donation) to avoid the “risk” of cardiovascular incidents. This often causes more harm than good, depleting the body of nutrients such as iron and B vitamins, while creating undue stress in replenishing the body's supply of blood.*

Elevated hematocrit is often due to dehydration and excess body fat, and can often be resolved by drinking more water before the test (people are often dehydrated for morning fasted blood work). I also notice trends towards higher hematocrit levels in clients with a higher body fat %.

Many reference ranges also only go to 0.5, while a more accurate reference range will go to 0.53 - which is a big difference, as many men on TRT sit at 0.50, 0.51, 0.52 or 0.53. I would argue there is likely little concern up to 0.55 and even beyond that depending on the individual, but this is something to discuss with your practitioner based on your symptoms and history. I have never had a client needing a therapeutic phlebotomy due to TRT.

What to expect:

In my experience, assuming there are no other health factors at play, this is where most TRT patients start “feeling” the majority of the benefits of TRT, particularly those who have been looking after themselves well.

*In my experience, TRT continues to snowball in benefits over a period of years (I am still noticing growing benefits after 6 years) as the majority of the benefits **begin** to “kick” in around the 6 month mark.*

At this point, it is also reasonable to assess your dose, to decide whether it is worth trying a higher or lower dose.

If you are noticing you are “hanging out” for your next injection and feeling significantly different before a dose vs after, I recommend increasing your injection frequency to the point where you can “forget” to take your dose and not notice any difference.

I recommend the dose schedules of twice a week, three times a week (Monday Wednesday and Friday) or daily.

? Why not every other day?

EOD becomes tedious over time as the days of the week are never the same, and with multiple half-lives banked, the two day break over the weekend is unlikely to cause issues for the majority of people on MWF protocols.

This makes it easy, allows you to build a routine, and means you don't need to worry about it over the weekend.

Note: For questions around dosing, adjustment amounts and target levels, please see the FAQs.

If things are feeling good at this point, I recommend a trial of HCG (discussed in Dialing It In) based on your goals regarding fertility (also discussed later).

If things need adjusting, I recommend making the adjustment and saving HCG for the next review. Changing one variable at a time allows for fewer side effects and more accurate outcomes. Remember, dialing in HRT is a marathon not a sprint, and your body takes time to adjust to things, so don't rush the process.

5 THIRD REVIEW AND BEYOND

I recommend waiting another 4-6 months after this appointment for another review to make sure everything is tracking along as previously, or to check the outcomes of any adjustments made at the previous appointment. This is the point where you are looking to get a "set and forget" protocol, where both you and your provider are happy with your blood work, your symptoms and your protocol.

If HCG has not been trialed, this is the time to introduce it into the protocol (discussed next).

After being on TRT for a year (and having the dose dialed in for at least half of that year) is when you should be feeling the benefits of testosterone replacement therapy.

? What are these benefits?

- *More energy*
- *Less anxiety and fear*
- *Improved mood*
- *Improved metabolism and appetite*
- *Body recomposition (depending on diet and exercise)*
- *Improved immune system*
- *Improved insulin sensitivity*
- *Improved sleep*
- *Improved cognitive function*

In my opinion, the greatest benefit of TRT at this point is **masculinisation** due to the androgenic traits of TRT.

This is the “psychoactive” effect of testosterone, which I believe is the benefit of testosterone that cannot be achieved by any other compound.

Men become more confident, less timid and more driven towards their goals due to the direct effect testosterone is having on their brain.

What you do with this is up to you and your responsibility.

An important finding I've had from working with many clients in different circumstances over the last two years is if you want to put the optimal amount of testosterone into your body that it COULD naturally make, you have to put your body in the state where it would be making these levels naturally, if it weren't for TRT.

This means that if you're overweight, not exercising or eating well, or trashing your body with poor lifestyle, you will be far more prone to the side effects of TRT, as your body will naturally be suppressing testosterone production under these circumstances. We must work with our bodies, not against them.

CHAPTER IV
DIALING IT IN

Getting “dialed in” is one of the main reasons men are seeking help with their TRT.

This may be because they have initiated treatment with an uninformed provider, or they are self-medicating and have gotten themselves halfway there, but have not resolved all of their negative symptoms.

In my experience, the biggest factor to getting dialed in is your diet and lifestyle. *There is no magic amount of mg of testosterone per shot that will make you feel great on a diet of processed food and a sedentary lifestyle although many people are looking for this.*

In fact, if your testosterone levels are at least 80% of the way to where they should be, the dialing in process should be taking you from “great” to “excellent” - the biggest movers are the kilograms of food you put into your body, not the difference in a few mg of testosterone injected per shot - trust me on that. There are no shortcuts.

I have also seen many guys trying to solve all their health problems with the perfect testosterone dose.

Unfortunately, when providers aren't aware of other hormones, or other health issues, they will try to solve all your problems with tweaking your testosterone dose and there is no magic dose of testosterone that will fix a sluggish thyroid or a deficiency in pregnenolone or DHEA.

If you picked up issues in your thyroid or other hormones from your preliminary blood work, I suggest working on those with a similar process to optimising your testosterone levels. This process is covered in “Beyond TRT”.

WHAT LEVEL SHOULD I AIM FOR?

This is one of the most commonly asked questions in TRT communities I have joined.

And while the “correct” answer is “there is no perfect level for everyone”, and that one person may feel great at x and another at y, we do need some rough guidelines to aim for.

When initiating treatment, I like to bring men to a free testosterone of around 600-1000 pmol/L. I will aim a little lower for older men and a little higher for younger men. *In general, I do find that younger men do better with higher levels and older men do better with lower levels, on a spectrum of optimal testosterone.*

There is a myth that TRT is about restoring the levels of testosterone to that of a healthy young man. *I look at TRT as restoring the optimal amount of testosterone for YOUR age ... and that amount is going to be drastically different at 21 or 80.*

I do not believe a man would tolerate the optimal dose for himself at 21, at 80 years of age, and would likely become overstimulated and experience side effects. *However, there will always be exceptions, as neurochemistry genetics are at play, as well as sensitivity and availability of androgen receptors, that are predominantly unknown factors for most people (unless a thorough genetic analysis has been done, but even this has limitations).*

This is why I like to aim for this level as a “ballpark” and titrate up or down based on symptoms.

The lowest free testosterone I have seen for a man who was “dialed in” is 550 pmol/L. I have seen a few outliers higher, but the highest free testosterone I tend to see for guys to get dialed in on is around 1500 pmol/L.

When explaining it simply, I say shoot for the top of the reference range play somewhere within the top of the range and double the top of the range. I have said time and time again that if you’ve got negative symptoms and a free testosterone of 1500 pmol/L doesn’t solve it, you’ve got other issues that need work.

Personally, I do best with a free testosterone around 1300-1500 pmol/L.

I have seen some outlier men who even have free testosterone levels in this range naturally (yes, it is possible, no is not supra physiological; yes, it is rare).

The majority of my clients sit around 900-1200pmol/L. I see very few clients who do well with a free testosterone under 700. I see a decent handful of clients who start to get negative side effects when their free testosterone goes above 1100-1200pmol/L, but I see equally as many who don't receive much benefit from treatment until their free testosterone level goes above 1200pmol/L.

I see most men over 60 do best with a free testosterone between 600 and 900 pmol/L. *(Many men over 60 also need thyroid, pregnenolone and DHEA. These deficiencies cause issues when testosterone is optimised and they are deficient.)*

I see most men under 25 do best with a free testosterone between 1200 pmol/L and 1500.

HOW MUCH SHOULD I ADJUST MY TESTOSTERONE DOSE?

Unless you are way off the mark, I recommend adjusting the dose in increments of no more than 25mg.

Although it may not sound like a lot, even 10mg a week of testosterone difference can be significant in how a man feels on TRT. The most I would adjust a TRT dose at one time is 50mg and that would be for someone being a significantly hyper or hypo responder at their first appointment.

The biggest issue (where an enormous number of men go wrong, especially in forums and communities) is men adjusting their dose based on how they feel that day.

Men - I want to make this clear.

You must be strategic and calculated with your protocol adjustments.

Throwing out your practitioner's advice and not following protocol because you feel a certain way on a certain day is not acceptable and failure to follow direction.

Dose adjustments should not be made without blood work and a full conversation with your provider.

Many men will constantly increase or decrease their dose based on how they feel (problem: your testosterone dose is far from the only thing that influences your mood and emotions) and do not stick with one dose for long enough to get an accurate blood reading.

Remember Testosterone Enanthate or Cypionate take 6-8 weeks to stabilise in the body.

I recommend adjusting your dose in doses of 10-25mg per week at a time, and getting blood work done every 8-12 weeks ideally before making further changes as the subjective effects of a certain dose take longer to achieve than stable serum levels.

It is normal to realise that optimal for YOU is a bit higher or lower than initially thought, once your body further adjusts to TRT, so don't be surprised if you review or change your dose at your 6 monthly or 12 monthly follow up with your provider.

I believe it is fundamentally important to optimise one's testosterone dose and do not leave potential benefits on the table. Do not be afraid to query your provider to request a trial of a mild dose increase with follow up bloods to assess and monitor safety and efficacy. The only way to find out if there is room and benefit to a dose increase (or decrease) is to try it and monitor symptoms and blood work, and adjust titrate as needed.

COMPOUNDED TESTOSTERONE CREAM

Compounded Testosterone Cream is a fantastic option for TRT.

But it is not my preference for a number of reasons.

Firstly, the response rate to cream is not as good as injectables. Many people seem to have issue absorbing the cream. This can come down to genetic issues (we assume) where the body simply does not respond to this application method.

There are also issues where the quality of the cream is not sufficient, as 20% testosterone cream requires a specific formulation to be fully effective. *Either a liposomal or atrevis base is needed to be specified to the compounding pharmacy, as this will enhance testosterone's absorption to allow it into the blood stream.*

Most providers recommend applying testosterone topically to the shoulders, inner arm/inner elbow or back of the knee. This is ineffective and increases risk of transference.

Testosterone cream is significantly more bioavailable when applied to a shaved scrotum, estimated to be up to 8x.

This is because the skin is thinner and therefore more testosterone makes it into the blood stream.

There are also theories that the testosterone works "better" because it is being applied locally to an area where it is made. However, given testosterone is systemic when it makes it into the blood stream, this is an interesting theory but is nothing more.

Another downside to cream is that it has a very short half-life. It is not clear exactly what the half-life is, as the duration of action is delayed by time taken to absorb the cream, as once the testosterone enters the blood stream, it is not bound to an ester and therefore has a half-life of minutes.

According to the research on scrotal testosterone cream application, 24 hours after application, testosterone concentrations are at around 50% of where they were at peak within approximately two hours after application.

Based on this, I believe twice daily application of cream is optimal, as a swing of 50% is a significant variance.

However, the main reason I recommend twice day application comes down to the practical realities of treatment.

If you apply your cream once a day and you forget to apply your dose (it will happen, trust me), you will be hypogonadal by the end of the day. *This twice daily application allows for human error (forgetting) without having a bad day.*

However, regular injections are a lot more forgiving, due to the multiple accumulated half-lives of the testosterone in the system. This means that when "life" happens and you cannot adhere to your dosing schedule, you will notice a very mild drop off and will take around 10 days to reach 50% levels - where you would be 24 hours after applying cream.

Not being able to access cream for multiple days in a row on cream is problematic, so cream is only suited to certain lifestyles. It does come with the benefit of not having to inject which is great and many men "feel" better on cream. *This is believed to be due to the higher conversion to DHT due to more 5AR enzyme (converts testosterone into DHT) being present in the skin.*

I believe this is also due to the large spike in free testosterone.

Due to these reasons, I recommend not starting with cream, but if you are interested in giving it a try, swap over to it once you are dialed in. That way, if you have issues, you'll be able to go back to the tried and true.

If you are going to use cream I recommend a starting dose of 0.25ml-0.5ml twice daily of 20% cream applied to the scrotum twice daily.

Most practitioners, including myself, measure bloods at around 4 hours after application to measure the peak and adjust up or down accordingly. It is normal to peak quite high (around 1500 pmol/l free testosterone) as there is a substantial drop over the 12- hour period prior to the following application.

Cream is also a good option for people who simply cannot tolerate injections (it happens) or for people who physically cannot inject due to injury or other health concerns.

HCG

HCG is one of the most misunderstood compounds in TRT.

There are two main questions to ask.

 Is it beneficial?

 Is it needed?

The answer to the first is, it depends on the individual.

The answer to the second is, no.

HCG is not required on TRT.

Many men all over the world thrive on TRT without HCG and men have been using testosterone for years prior to HCG being used in the application of TRT.

HCG binds to the LH receptor and may be beneficial as LH is reduced to zero in TRT. As it serves many of the same functions as LH, it is touted that LH “mimics” HCG.

This term is ambiguous.

HCG binds to the same receptors as LH, but it does not exert the same effect as LH systemically.

If it did, the female body (where HCG comes from) would not make an abundance of HCG during the first trimester of pregnancy, it would simply make more LH.

It is very well known that many men do not do well with HCG, while others do. I believe the difference between HCG and LH is the reason why some men do not respond well.

HCG's differences to LH are not well researched, but it is not known in the TRT community exactly why and who will respond to HCG well, and who will respond poorly.

- ❓ We know LH receptors are systemic throughout the body (not just located in the testicles) and the question is, is their activation required for unique functions, or beneficial to the human body, in the presence of exogenous testosterone? (The primary function of LH is currently understood as manufacturing sex hormones, which is already taken care of.)

The answer is, we don't know.

This is why I recommend men try HCG within the first year of their treatment irrespective of their fertility goals. *If you do well with HCG keep using it.* If you don't, then see the next section regarding fertility options. But one thing is for sure, it is best not to use HCG if it causes side effects.

The most common side effects are what men complain are "estrogenic" symptoms, yet they do not subside with aromatase inhibitor use. Some believe this is due to intratesticular aromatisation not being impacted by aromatase inhibitors (which is true), but many identify these negative symptoms as unique.

This is why it is important not to initiate TRT with HCG (*unless under specific circumstances for fertility*), as while 100% of biological men respond well to testosterone (*as it is vital to male health*), it is a coin toss whether an individual will respond to HCG.

This means it is very difficult for the practitioner to ascertain what side effects are coming from where.

Men need to establish a solid baseline of resolution of their hypogonadism symptoms, so that they can make an accurate assessment as to whether HCG makes them feel better or worse.

For HCG to be added to TRT, I prefer dosing of 50-150IU daily.

Some men do better on a lower dose, some need a higher dose to have their testicles respond fully.

Intratesticular testosterone production appears to be maximised at 250IU, and HCG has a half-life of around a day. For this reason, I only recommend HCG dosing either daily, or three times weekly/EOD at 250-300IU. 500IU 2x weekly may work for some, but most will notice fluctuations.

If you don't respond well to HCG and still want to use it as an insurance policy for fertility, I personally use 3x 500IU doses of HCG over a week, 3-4 times per year. This allows me to completely activate the leydig cells for a short period to "keep the signal" there. We do not know if this is necessary and this will likely never be studied. But, as someone who may not have children for a decade, I am more comfortable doing this than leaving my testicles dormant for 10 years.

Many men have had no issues conceiving after not using HCG for decades, but it is the individual's decision on whether they want to roll the dice.

SUBCUTANEOUS INJECTIONS

Injecting testosterone subcutaneously offers multiple benefits, but with a few caveats.

It is much more comfortable to inject into the fat than the muscle, especially over a period of years.

This also allows more comfort and better compliance with more frequent injection protocols, especially for men who benefit from daily testosterone injections.

Subcutaneous injections (injected into the subcutaneous layer of fat, rather than the muscle) requires the testosterone to be drained through the lymphatic system to be absorbed, that is a slower process, resulting in a lower peak and smoothing out the release further.

There are studies often cited in forums showing the effectiveness and advantages of subcutaneous injections.

However, there are an equal number of posts regarding men who cannot tolerate subcutaneous injections (subq) as it causes painful lumps and swelling, and a resurgence of hypogonadism symptoms. They say they feel like their testosterone “isn’t working”.

For the ones who have negative side effects and stick it out to get blood work before abandoning the cause, they will often have a reduction in serum levels by 30-50%.

It is important to note, and is often overlooked, that only one preparation of testosterone was studied for subcutaneous administration.

In my experimentation over the years, I have found the carrier oil is the determining factor for whether a testosterone formulation will work subcutaneously for most people.

The oils that seem to work best are sesame, grape-seed and mygliol (pharmaceutical grade mct). The study used sesame oil.

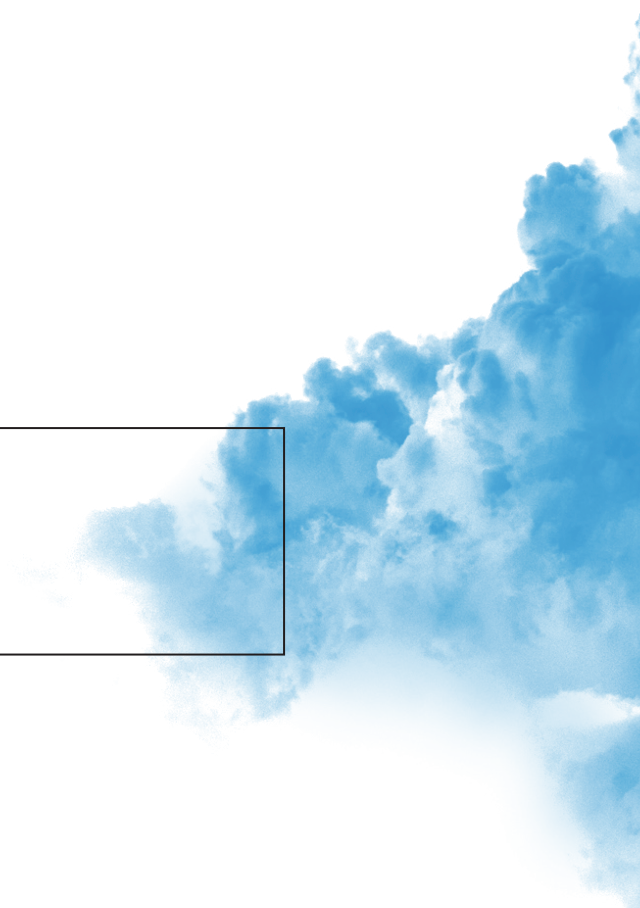
Oils that do not work for the majority of people are cotton seed, castor oil and peanut oil.

I still recommend beginning with intramuscular injections (IM) as this works for 100% of people, and is how the preparation is designed to be absorbed. Even using the approved subq oils some people simply do not respond as well as they do to IM.

(Oil goes into muscle, water goes into fat.)

Get your levels dialed in to the point that you've got a solid frame of reference, then with your provider's approval, swap to subcutaneous for 6-8 weeks, run a blood test and check your symptoms and levels.

If it works for you, great. If it doesn't, then stick with IM.



CHAPTER V

FAQs

? Does TRT cause gynecomastia?

Testosterone replacement therapy restoring optimal male androgen levels should not cause a man to grow female breast tissue.

However, if a TRT protocol is given with a large weekly or fortnightly bolus dose causing unnatural hormonal variances, (particularly with hypogonadal periods after a fortnight) then this is a possibility.

There are also some people who have a genetic predisposition to develop gynecomastia. They may have already developed in puberty and TRT aggravated it (increased hormones will do this). Others will have it occur in later life when they initiate TRT. It is important to understand that these people with genetic predispositions will have developed gynecomastia had they had optimal, natural testosterone levels.

The solution is to have the gland surgically removed.

Using SERMs (tamoxifen, clomid) to block the receptors in the breast will only work while the drug is present in the body and taking SERMs long term causes problems with the liver and brain, and is unwise and not well tolerated. Some men say that using tamoxifen for a period can get rid of gynecomastia, even when it developed during puberty. I have found that the gyno does end up coming back and inevitably needs to be surgically removed.

? Can I use Sustanon?

Sustanon is often the preferred choice of testosterone for men in the UK, as it is readily available and cheap.

Sustanon has a blend of 4 esters, two fast and two long acting, designed to deliver stable release of testosterone over a longer period than enanthate or cypionate.

This looks and sounds great on paper, but, in reality it doesn't work well for infrequent administration (often it is recommended every 3 weeks), as men feel inconsistent due to the varying peaks and valleys of multiple esters.

I find Sustanon works great for daily, shallow IM injections (Sustanon doesn't work well subcutaneously for the majority of men due to the peanut oil) as men enjoy the daily spike of the propionate and phenyl propionate, and then have stable, long acting esters in the background.

But, unless you want to inject daily, Sustanon is going to deliver less favourable outcomes for the majority of men than enanthate or cypionate.

Will my balls shrink?

Yes.

Most men notice a reduction in testicle weight and size by an average of 30%. Given this 30% is average, it means there will be some who have more and others who have less.

Most men are not bothered by the testicular shrinkage as it is mild and doesn't affect the function of the penis or orgasm.

If you have issues with the testicles pulling up into the body, I recommend strengthening the pelvic floor and hips, as this often resolves the issue.

HCG will restore testicular size for most men (full size or close).

Some men feel bad on HCG but are bothered by the smaller testicles. I generally recommend they work on accepting the smaller testicles.

? Will my penis shrink?

No. The vast majority of men report fully, better, stronger erections on TRT.

Androgens (particular DHT) are responsible for penis growth during puberty and some men report their penis growing on TRT. However, TRT is not known for growing the penis consistently.

? What if I want to stop TRT?

I do not recommend people stop TRT abruptly without a Post Cycle Therapy protocol (PCT) beyond the first month.

PCT protocols generally include a SERM and HCG for a 4 week period after cessation of TRT to restart the body's natural production. PCT protocols are discussed in depth in Beyond TRT.

? Can I do a trial of TRT?

I do not advise people frame TRT this way.

If TRT causes unmanageable side effects or must be ceased for extenuating circumstances, a PCT can be performed to regain the body's natural production. However, there is no guarantee the body will regain 100% of its natural production and a PCT is stressful on the body (SERMS are nasty drugs).

TRT can be withdrawn (ideally in the first 6 months) if there are problems, but I encourage people to begin TRT under the assumption that it is for life.

It would be great if we could do a risk-free trial of TRT, but the male body does not allow it.

? Can I drink alcohol on TRT?

Yes, but the health of the liver is extremely important in the outcomes of TRT.

Many men have side effects on TRT due to liver issues.

I do not recommend binge drinking of alcohol period. However, alcohol in small doses will not have any worse outcomes on a man on TRT as opposed to a man not on TRT.

? Who are the best TRT providers?

While I am sure there are many other great providers of TRT in the world, the vast majority of doctors prescribing TRT have very little understanding of how testosterone works, and run a risk of doing more harm than good in working with men on TRT.

Unless you are aware of a doctor with an excellent track record and reviews, I recommend seeking out one of the following providers:

- Justin Groce - The Restore Clinic (US)
- Elevate Men's Clinic (US)
- Dr Jordan Grant (US)
- Dr Jeffrey Ruterbusch (US)
- Dr Jim Meehan (US)
- Dr Eric Serrano (US)
- Balance My Hormones (UK + Europe)
- The Mojo Clinic (UK)

? What are the best sources of information on TRT?

I recommend the following resources:

- Danny Bossa: Youtube
- Jay Campbell: The TOT Bible + Youtube/Podcasts
- TRT And Hormone Optimization Youtube Channel
- Dr Neal Rouzier - World-link Medical Conferences

? Will TRT put on muscle?

Optimising testosterone levels will cause some degree of muscle growth, but how noticeable that will be will depend on diet and exercise.

TRT will give you the results you deserve from the work you put in, while having low testosterone will generally cause the body not to respond properly to strength and hypertrophy training.

If you've previously been discouraged from weight training or exercise in a low testosterone state, I highly recommend revisiting training with a proper workout plan featuring progressive overload and a good nutrition regime.

? Will TRT cause heart disease?

No. Optimal testosterone levels do not cause heart disease.

A poorly managed TRT protocol with large bolus doses can cause inflammation and prolonged hypogonadal states (before the next shot is due) and can contribute to heart disease.

Testosterone is cardio-protective. This is why heart disease tends to occur later in life, not when testosterone levels are peaking in youth. Testosterone is protective against many diseases of ageing.

? Will TRT cause prostate cancer?

No.

Low testosterone levels for prolonged periods have been shown to cause BPH and prostate cancer.

It is generally out of shape, elderly men who develop prostate problems, not young healthy athletic men with optimal testosterone.

There are some doctors who even advocate treating prostate cancer with TRT.

? Will TRT make me lose my hair?

Yes and no.

If you are predisposed to male pattern baldness, then balding is inevitable if you want optimal testosterone levels, as male pattern baldness is due to androgen receptor activation in the scalp.

If you do not have the genes for male pattern baldness, TRT will not cause hair loss.

? Can I take Finasteride or Dutasteride to keep my hair?

I highly recommend people do not touch these compounds in any form.

While there are proponents of its use with TRT, I consistently see men with PFS (Post Finasteride Syndrome) with devastating damage to their physical and mental health. It is well known that the side effects of finasteride are up to 15% in most studies, and there was a class action against the manufacturer where it was deemed that the side effects could indeed be permanent, and the manufacturer was aware from the clinical trials.

Heard different elsewhere? Just go to the finasteride wikipedia, you can link to the meta-analysis and the transcript and read them yourself.

DHT (that is what is reduced by finasteride) is responsible for the androgenic (masculinising) effects of testosterone and antagonises the effects of oestrogens' (including xenoestrogens). DHT, as well as allopregnenolone (also blocked by these drugs) are two of the most crucial neurosteroids for man's brain and impacts mental health as well as preventing cognitive decline long term. *It is not worth risking all aspects of your health and masculinity to keep a pretty head of hair.*

You can become more attractive through developing your physique and a man's value comes from what he does, not how pretty his haircut is.

DHT is one of the main benefits of TRT for mood and DHT derivatives (androgens similar to DHT like proviron) can even be used in therapeutic doses to treat depression and libido issues as an adjunct to TRT (discussed in Beyond TRT).

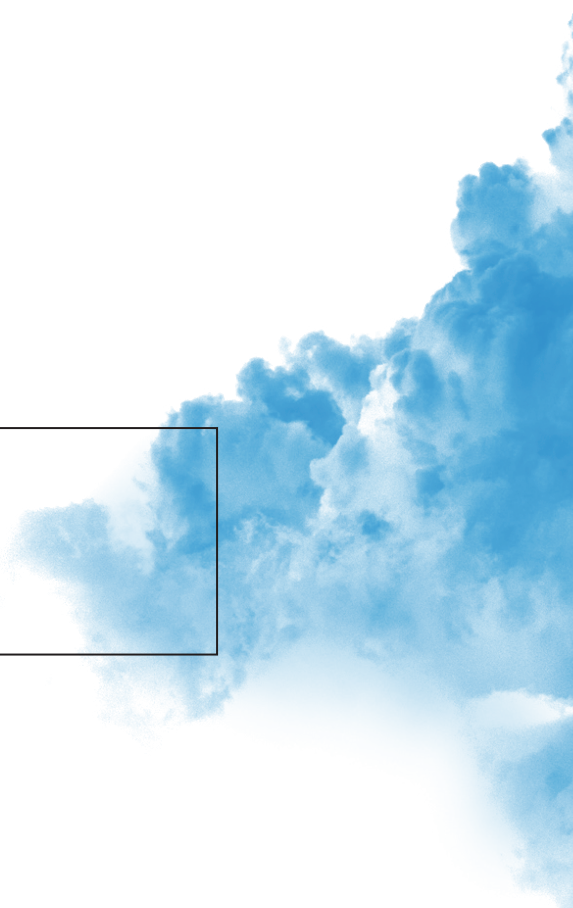
Should I take any supplements with TRT?

I don't believe TRT indicates any specific supplements, but there are a variety of supplements that can be beneficial to augment TRT.

Most supplements that are beneficial are individualised for specific purposes (discussed in Beyond TRT).

The 3 supplements I recommend for everyone on TRT are Basic Nutrients by Thorne (Comprehensive Multi Vitamin that is actually bioavailable) CBD oil (reduces inflammation, reduces stress, antioxidant effects) and L Glutathione, in either an injectable form or via Thorne's SR product (Liver antioxidant).

These are elaborated further in Beyond TRT. To start, I'd pick up Basic Nutrients by Thorne.



CHAPTER VI

AFTERWORD

This guide is dedicated to the memory and work of the late Dr John Crisler, who pioneered many practices now considered mainstream in TRT.

I sincerely hope this guide brought you value and a further understanding of your journey on TRT. As someone who has been personally using TRT for over 5 years, and having coached over 500 men, working closely with them on all aspects of their physical and mental health, I truly believe that testosterone optimisation is the future of men's health. The transformations I see not only physically, but in character, frame and mental point of origin, in the men I work with is nothing short of amazing, and goes to show how integral testosterone is to male biology.

I believe we are going to see a significant increase in cases of hypogonadism in young men and I truly believe one of the primary underlying root causes of mental health problems in both young men and women alike is endocrine disruption. There are hundreds of thousands of men who are unknowingly suffering with hypogonadism, often diagnosed with labels such as "depression" or "anxiety" and doctors are attempting to use psychiatric drugs to treat these symptoms, that are just causing worse physical and mental health outcomes over time.

If TRT changes your life, I strongly implore you to be an ambassador for its use, and bring other men up with you to achieve their masculine potential. Raising awareness of the testosterone deficiency epidemic on a grassroots level can have far reaching benefits in getting men to get their levels tested, especially for those who aren't doing well, and where their doctors dismissed their levels and symptoms as "normal". In a world where "normal" is sick, depressed and overweight, I congratulate you in seeking the cure for your ailments and uncovering the solution to your own problems.

If you enjoyed this guide and would like to work with me further or engage with more of my content, you can find links on:



[Advanced Fundamental Health - http://www.advancedfundamentalhealth.com](http://www.advancedfundamentalhealth.com)

Best Regards,
Dave

Copyright © 2022 TRT 101 by Dave Lee.

All rights reserved. No part of this book may be reproduced or used in any manner without the prior written permission of the copyright owner, except for the use of brief quotations in a book review.

DISCLAIMER:

The information contained on this book is provided for educational purposes only. It is not meant to replace professional medical advice, diagnosis, or treatment. Any attempt to diagnose and treat a medical condition should be done under the direction of a healthcare provider or physician. For any medical conditions, each individual is recommended to consult with a healthcare provider before using any information, idea, or products discussed. Neither the authors nor the publisher shall be liable or responsible for any loss or adverse effects allegedly arising from any information or suggestion on this book. While every effort has been made to ensure the accuracy of the information presented, neither the authors nor the publisher assumes any responsibility for errors. References are provided for information purposes and do not constitute endorsement of any websites or other sources. Readers should be aware that the websites listed here may change.

ISBN: 979-8-88722-128-1 (e-book)

Cover and layout by Aliel Kika.

First Edition.



www.advancedfundamentalhealth.com

"David has been instrumental in not only helping me to recover from a chronic injury but also in getting me into the best condition of my life, mentally and physically. A genuine, no-nonsense, knowledgeable practitioner who cares about his clients. He has helped with recovery, training, diet, supplementation, mobility, hormone balancing, stress management, sleep, and mental wellbeing for a holistic approach to health, which is all too rare. Highly recommended."

— Isaac S, NSW Australia

"I've been lost and frustrated for years dealing with endocrinologists and doctors. My health, energy, mood and weight had declined significantly during that time and I was sick of my concerns being brushed aside by the medical profession as "something I should expect as I get older". After lots of searching for answers I came across Dave and I'm so glad I did. At long last I've found someone who listens and genuinely cares about my health and has my best interest at heart. Dave is super generous with his time, super knowledgeable and is more than happy to share that knowledge with his clients. Best of all, he knows how to get results. Finally I've found somebody I can trust."

— Ricky, NSW Australia

"I reached out to Dave as I had been managing my own TRT and whilst I was doing ok, I was not managing the main problems of my mental health suffering from ADHD and depression. After a couple of deep conversations with Dave, I could see that he actually cared for my wellbeing. 8 months since our first consultation, I am now happier, more confident and feeling amazing following his supplement protocol and lifestyle change advice"

— Brett, WA Australia



TESTOSTERONE REPLACEMENT
THERAPY - 101 BY DAVE LEE

"I have no idea why Dave Lee is not a household name yet, but it's coming, I can assure you. So much so that, while he is living in Australia, and I in Canada, I stopped counting how many family members and friends I have sent to this masterful individual with a wealth of knowledge that encompasses the areas of health, fitness, medicine, diet, supplementation, and more. Most physicians follow the 'sick care' philosophy but will do virtually nothing to help you maintain, or better yet improve your health in order for you to live your best life. This is where Dave Lee is virtually unmatched as his skillset encompasses such a vast array of specialities that make most physicians look completely one dimensional. If you have followed me on the TRT and Hormone Optimization YouTube channel you'll know that I'm straight, to the point, and no BS. I can absolutely recommend Dave Lee, hands down, as being in my top picks, worldwide, as well as someone who has made a genuine and measurable impact on my life and the lives of everyone I have referred to him. I will continue to recommend this wonderful young man and wish him nothing but success as well as continued growth of knowledge and undeniable passion in this field. Thank YOU Dave!"

— Danny Bossa, Montreal, Canada

"I suffer from several complex health issues, and began working with Dave in Early 2019 after getting nowhere with various doctors and health practitioners. Dave has demonstrated immeasurable expertise, patience and care, and has truly changed my life for the better in many ways. His science-based, holistic approach to health has opened a whole new life for me, and helped me to optimise my hormones, live to the advantage of my genetics, identify food intolerances and improve my relationship with food, improve my gym training, mental health, confidence, personal relationships and my chronic pain condition. I feel like a new person since working with Dave. My only regret is not investing in his services earlier. I really appreciate all that he has done. I'm grateful for Dave thinking of me as a whole person, his encouragement and his patience."

— Chris, TAS Australia

